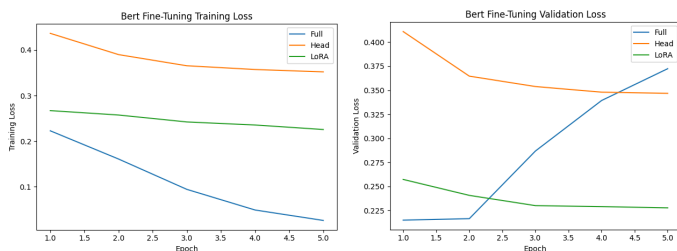


CS190I Programming Assignment 2

Simon Yan

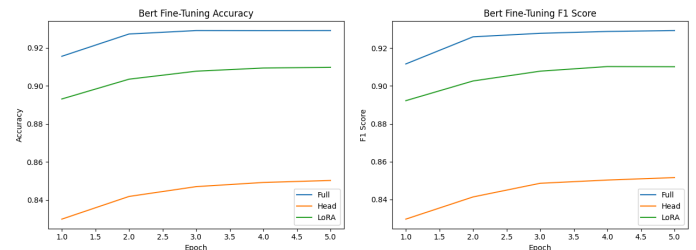
I will be fine-tuning DistilBERT using the IMDB movie review dataset. DistilBERT is a smaller model of Huggingface's BERT with half of the parameters, from 110M parameters to 66M parameters, and from 12 layers to 6 layers. DistilBERT offers 97% of BERT's accuracy, making it a leading option for faster inference and smaller model size without compromising too much model computational accuracy. The IMDB movie review dataset is a sentiment analysis dataset of 50,000 reviews, which are classified as positive or negative. In my project, I compare different methods of fine-tuning the DistilBERT on the IMDB movie reviews dataset.

I use three different methods of fine-tuning. The first method is full fine-tuning, in which all of the parameters and weights of the DistilBERT model are updated in training. The second method is head training, which unlike full training, freezes all layer parameters and weights and only updates an output layer. The third method is LoRA, or low-rank adaptation, which keeps the original parameter weights but adds small matrix layers that are trained to fine-tune the model to predict the given dataset.



As you can see in the training, full fine-tuning provides the least loss in training. This is because all of the parameters are being updated. However, in validation loss, there are obvious signs of overfitting, meaning that updating all 66M parameters and weights of the DistilBERT model is too much computational power for the IMDB movie

reviews dataset. LoRA, on the other hand, has slightly more loss but in 5 epochs, the training loss and validation loss both are decreasing. For head only training, the loss is much higher than the previous two methods, making it not an ideal fine-tuning method in this case.



The accuracy and F1 score show these same trends as well. Although the full fine-tuning method saw some overfitting in the training and validation sets, it provides the best results in terms of accuracy and F1 score on the test dataset. Head-only training is much less accurate than full fine-tuning and LoRA. In terms of training time, head-only fine-tuning took 56 minutes and 30 seconds to train 5 full epochs. LoRA fine-tuning took 1 hour 43 minutes and 12 seconds, while full fine-tuning took 2 hours 12 minutes and 58 seconds. The trade-offs between accuracy and time for head-tuning are easily identifiable as not ideal as you get much lower accuracy. However, between full fine-tuning and LoRA, the trade-offs can be debated. Either you choose LoRA for ~20% faster training and lose ~2% accuracy or you choose full fine-tuning with longer training time and better accuracy, but slight overfitting in even a reduced-sized DistilBERT model.

References

https://colab.research.google.com/drive/14DOeKIJ_6deeacVK0thyDEMaup0Ukkij?usp=sharing
https://huggingface.co/docs/transformers/en/model_doc/distilbert
<https://github.com/huggingface/peft>
<https://huggingface.co/docs/transformers/v4.14.1/en/training#finetuning-in-pytorch-with-the-trainer-api>